Tomato Disease Management

Covers: Early Blight, Septoria Leaf Spot, Late Blight, Bacterial Spot/Speck/Canker, Leaf Mold, Healthy

Compiled from public university cooperative extension publications for testing InsightAI-RAG's retrieval and grounding quality against real agricultural guidance. This is not a substitute for consulting a local agronomist or extension office before applying any treatment. Always follow current product labels; pesticide registrations change and vary by region.

Early Blight — Alternaria linariae (formerly A. tomatephila / A. solani)

Lesions on established plants appear as small, brown-black spots on older foliage, especially where leaves touch the soil; they enlarge rapidly with concentric target-like rings, surrounding tissue yellows, and overall blighting and defoliation follow. Fruit infections develop near the stem attachment, brown-to-black with concentric rings; a thick mass of black spores develops under humid conditions. Optimum infection temperature is 82-86F, but lesions can appear about 7 days after infection even under less favorable conditions. Spread is mainly by wind and rain splash, plus human/equipment contact.

Management: Rotate away from tomato, potato and other nightshades for at least 2 years. Choose tolerant varieties (e.g. 'Iron Lady F1', 'Mountain Merit F1', 'Defiant F1' have partial tolerance). Reduce leaf wetness duration via wider spacing, row orientation with prevailing wind, morning sun/airflow, and avoiding overhead irrigation. Mulch under plants to reduce soil contact. Clean equipment and remove/bury plant debris at season end. Numerous fungicides are labeled (FRAC groups 3, 7, 9, 11, 27 and combinations); rotate by FRAC code and consult a current regional guide, since some populations have developed FRAC 11 resistance.

Source: University of Kentucky Plant Pathology Fact Sheet PPFS-VG-25 (Dec 2020); University of Tennessee Extension SP277-W 'Foliar Diseases of Tomato'

Septoria Leaf Spot — Septoria lycopersici

Develops first on lower foliage after first fruit set and progresses up the plant. Circular-to-semicircular spots have tan-to-gray centers with dark brown margins, often with a yellow halo; centers become dotted with tiny black fungal fruiting bodies (pycnidia), the key distinguishing sign versus early blight. Individual spots stay under 1/3 inch but often merge into a blight that defoliates the plant. Fruit infection is rare. Symptoms can appear within 6 days of infection; the fungus persists on plant debris, weeds, and even wood/metal stakes and cages between seasons.

Management: Same cultural practices as early blight (rotation, spacing, avoiding overhead irrigation, sanitation) are effective, except no Septoria-resistant varieties currently exist. Fungicides labeled for early blight generally also control Septoria — see the Early Blight entry's FRAC guidance.

Source: University of Kentucky Plant Pathology Fact Sheet PPFS-VG-25 (Dec 2020); University of Tennessee Extension SP277-W 'Foliar Diseases of Tomato'

Late Blight — Phytophthora infestans

Sporadic but potentially devastating and fast-spreading, requiring mild, moist weather. Water-soaked lesions enlarge into irregular greenish-black blotches that dry and resemble frost damage, often with a pale green halo; white, downy fungal growth appears on the underside. Fruit develops large, irregular brown blotches leading to whole-fruit rot; stems show brown discoloration. Distinguishable from early blight by lighter tan lesions with a light green halo occurring on young leaves throughout the plant, not just the lower/older foliage. Inoculum sources include infected transplants, potato tubers, or wind-borne spores from other areas.

Management: The same spray programs and moisture-management practices used for early blight also help control late blight. Some varieties carry resistance to specific late blight strains. Source transplants from local, reputable greenhouses to avoid importing the disease.

Source: University of Tennessee Extension SP277-W 'Foliar Diseases of Tomato'; University of Wisconsin-Extension A4052-02 'Potato Late Blight' (same pathogen, cross-applicable biology)

Bacterial Spot, Speck & Canker — Xanthomonas spp. / Clavibacter michiganensis

Bacterial spot: small, dark, circular-to-irregular lesions easier to see on leaf undersides, more water-soaked when wet; scabby, sometimes sunken fruit lesions. **Bacterial speck** looks similar on leaves but produces smaller, shallower fruit lesions. **Bacterial canker** causes marginal leaf browning, wilting, whole-plant death, and longitudinal stem cankers that split open; fruit spots have dark raised centers with white halos. All three bacteria are commonly introduced on infected transplants and can survive on seed, in plant debris, and on tools; spread is via splashing rain, machinery, and workers' hands.

Management: These bacterial diseases are difficult to control once established, especially in frequent rain. Fixed copper products mixed with mancozeb may help. The most effective control is exclusion — source transplants from local, reputable greenhouses and avoid out-of-region stock.

Source: University of Tennessee Extension SP277-W 'Foliar Diseases of Tomato'

Leaf Mold — Fulvia fulva

Primarily a greenhouse problem, occasionally seen outdoors. Pale green or yellowish spots appear on the upper leaf surface with an olive-green-to-tan velvety fungal growth on the underside. Favored by long periods above 85% humidity; complete defoliation can occur under favorable conditions.

Management: Lower greenhouse humidity, plant resistant varieties, and apply fungicides promptly when conditions favor the disease.

Source: University of Tennessee Extension SP277-W 'Foliar Diseases of Tomato'

Healthy

No signs of disease detected. Maintain airflow, avoid overhead watering, and scout weekly, especially after wet weather.

Source: General extension guidance